

Smart Traffic Lights Using Local AI for Adaptive Traffic Control and Emergency Vehicle Priority

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Abstract—Urban traffic signals commonly rely on fixed timing plans, which may perform poorly when traffic demand changes rapidly. Recent research has investigated adaptive traffic signal control using machine vision, deep reinforcement learning, and edge computing. This paper proposes a local-AI traffic-light architecture in which roadside cameras provide traffic information to an edge computer, the controller adapts signal timing, and an emergency-vehicle module can temporarily coordinate signal priority.

The research gap is identified as the integration of local vision-based perception, adaptive signal timing, emergency priority, and recovery of normal traffic while limiting disruption to non-priority traffic. The proposed system can reduce dependence on continuous cloud communication by performing important processing close to the traffic intersection. The paper also defines an evaluation framework using vehicle delay, queue length, emergency-vehicle delay, recovery time, processing latency, and network-data dependence.

Index Terms—Smart traffic lights, Local AI, Edge computing, Machine vision, Adaptive traffic signal control, Emergency vehicle preemption, Intelligent transportation systems.

I. INTRODUCTION

Traffic congestion is strongly influenced by changing vehicle demand, intersection queues, and signal timing. Traditional traffic lights commonly operate using fixed timing plans. Although fixed-time signals are simple and inexpensive, they cannot directly respond to sudden changes in traffic conditions.

Artificial intelligence has increasingly been investigated for adaptive traffic signal control. Rasheed et al. reviewed deep reinforcement learning techniques for traffic signal control and identified AI-based approaches as an important research direction in intelligent transportation systems [1].

Machine vision provides another method for adaptive traffic control. Camera-based systems can obtain information such as vehicle counts and traffic flow and use this information to modify traffic signal timing [2].

Another important approach is edge computing. Instead of sending all camera information to a remote cloud server, some processing can be performed near the traffic intersection. This can reduce communication requirements and potentially improve response time [3].

Therefore, this research proposes a local-AI traffic-light architecture that combines traffic perception, adaptive signal control, emergency vehicle priority, and controlled recovery.

II. LITERATURE REVIEW

A. AI-Based Traffic Signal Control

Deep reinforcement learning has been widely investigated for traffic signal control. Rasheed et al. presented a review of deep reinforcement learning methods and discussed their application to traffic signal control [1].

These approaches demonstrate that AI can be used to make traffic control decisions based on changing traffic conditions rather than only using fixed timing.

B. Machine Vision

Gorodokin et al. investigated adaptive traffic-light control based on machine vision. Camera information can be used to estimate traffic conditions and support changes in signal timing [2].

This provides the basis for using roadside cameras as an input to a local traffic-control system.

C. Edge Computing

Edge computing places computation closer to the source of data. In traffic systems, edge-based processing can allow traffic information to be processed near the intersection instead of continuously sending all information to a central cloud server [3].

D. Emergency Vehicle Priority

Emergency vehicle signal preemption has also been investigated. Wang et al. proposed an emergency-vehicle-centered traffic signal control approach for intelligent transportation systems [4].

Chen et al. investigated emergency vehicle signal preemption while considering queue spillbacks and the negative effects on non-priority traffic [5].

These studies show that emergency vehicle priority is already an established research area.

III. RESEARCH GAP

The original motivation for this work is that conventional traffic systems may not automatically respond to emergency vehicles. However, existing research has already investigated emergency vehicle signal preemption [4], [5].

Therefore, the research gap should not be stated as the absence of emergency vehicle priority.

The research gap identified in this paper is the need for an integrated and practically evaluated architecture combining:

- Local camera-based traffic perception.
- Adaptive traffic signal timing.
- Emergency vehicle detection or priority input.
- Emergency signal coordination.
- Controlled recovery after emergency passage.
- Reduced dependence on continuous cloud communication.

The proposed architecture attempts to combine these functions into one local-AI framework.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed system consists of four major stages:

- 1) Street cameras capture traffic conditions.
- 2) A local AI/edge computer processes the camera information.
- 3) An adaptive signal controller decides the appropriate signal strategy.
- 4) Traffic lights execute the selected signal sequence.

An optional cloud server can be used for data storage, monitoring, and long-term traffic analysis. However, continuous cloud processing is not required for every traffic-control decision.

V. SYSTEM FLOWCHART

Figure 1 presents the proposed system flow.

VI. WORKING PRINCIPLE

The working principle of the proposed system is as follows.

First, cameras continuously observe traffic conditions around the intersection. The local AI system processes the camera information and estimates traffic conditions.

Second, the adaptive controller determines the required signal timing. If normal traffic is detected, the controller attempts to allocate signal time according to traffic demand.

If an emergency vehicle is detected or an authenticated emergency priority signal is received, the controller activates an emergency-priority strategy.

After the emergency vehicle passes through the intersection, the controller gradually restores normal signal operation.

VII. COMPARISON OF TRAFFIC CONTROL SYSTEMS

Table I presents a qualitative comparison of different traffic-control approaches.

The proposed system is intended to perform important decisions locally while allowing optional cloud communication for storage and city-level analysis.

VIII. LITERATURE-BASED RESEARCH GAP

Table II summarizes the research gap.

TABLE I
QUALITATIVE COMPARISON

System	Decision Basis	Internet Dependence
Fixed-Time	Preset clock	None
Cloud AI	Central processing	High
Local AI	Local traffic information	Low/Optional
Proposed System	Traffic density + emergency condition	Low/Optional

IX. EVALUATION FRAMEWORK

A simulation or prototype implementation should compare the proposed system with fixed-time traffic control and an adaptive non-emergency baseline.

The following performance measures can be used.

The experiment should include low-, medium-, and high-traffic conditions. Emergency and non-emergency scenarios should also be tested.

X. ADVANTAGES

The proposed local-AI architecture has several potential advantages:

- Faster local decision making.
- Reduced dependence on continuous internet connectivity.
- Adaptive response to changing traffic conditions.
- Support for emergency vehicle priority.
- Optional cloud connectivity for long-term analysis.
- Modular architecture that can be extended to multiple intersections.

XI. LIMITATIONS

Camera-based systems can be affected by lighting, weather, camera position, and vehicle occlusion.

Emergency vehicle detection must also be carefully validated because incorrect detection can affect traffic signal operation.

The proposed architecture is a research framework and should not be interpreted as evidence of completed real-world deployment.

XII. FUTURE WORK

Future work can investigate lightweight AI models suitable for edge hardware. More experiments can be conducted using realistic traffic datasets and traffic simulators.

Future studies can also investigate:

- Multi-intersection coordination.
- Privacy-preserving video processing.
- Emergency vehicle authentication.
- Advanced recovery strategies.
- Hardware-in-the-loop testing.
- Real-world intersection testing.

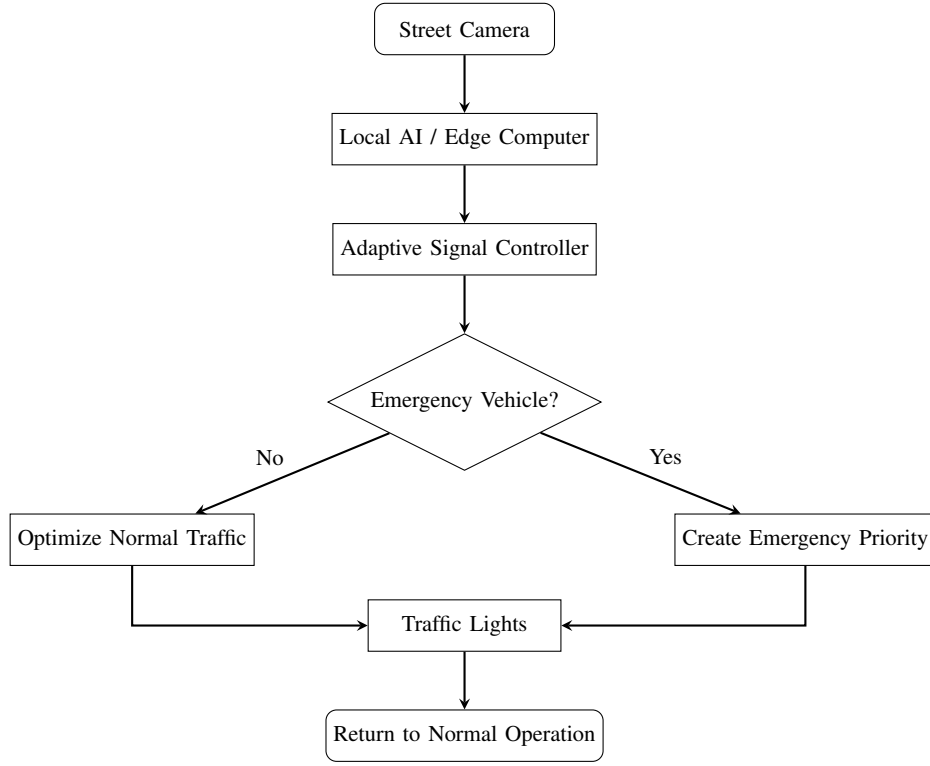


Fig. 1. Proposed local-AI traffic control flow with emergency priority and recovery.

TABLE II
LITERATURE-BASED RESEARCH GAP

Research Area	Existing Research	Remaining Integration Issue
AI Traffic Control	AI and deep reinforcement learning have been studied for traffic signal control.	Need practical integration with local traffic perception and emergency handling.
Machine Vision	Camera-based traffic information can support adaptive signal timing.	Need robust emergency-aware control under different traffic conditions.
Edge Computing	Traffic processing can be performed close to intersections.	Need integration with adaptive signal control and emergency priority.
Emergency Preemption	Emergency vehicle signal priority has already been investigated.	Need integration with local vision, adaptive traffic control, and recovery.
Traffic Recovery	Queue spillback and recovery after emergency priority have been studied.	Need a unified local-AI evaluation framework.

TABLE III
PROPOSED EVALUATION METRICS

Metric	Purpose
Average vehicle delay	Measures normal traffic efficiency.
Maximum queue length	Measures congestion buildup.
Emergency vehicle delay	Measures emergency-priority effectiveness.
Emergency travel time	Measures end-to-end emergency response.
Recovery time	Measures restoration of normal operation.
Processing latency	Measures local AI response time.
Network dependence	Measures the requirement for cloud communication.

XIII. CONCLUSION

This paper presented a local-AI architecture for adaptive traffic signal control and emergency vehicle priority.

Existing literature demonstrates that AI-based signal control, machine vision, edge computing, and emergency vehicle preemption are already established research areas. Therefore, the research gap is better defined as the integration and evaluation of these components in a practical low-latency architecture.

The proposed system uses roadside cameras, local AI processing, adaptive signal control, emergency priority, and controlled recovery. The evaluation framework provides measurable parameters such as vehicle delay, queue length, emergency vehicle delay, processing latency, and recovery time.

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